Attention Kernel Optimization Notes

SYNTHETIC TEST DOCUMENT. All values in this document were invented to test the OmniResearch software and do not report real experiments.

This report compares fused attention kernels. Page 2 is a scanned memo reproduced as an image. A separate benchmark found continuous batching improves throughput by only 8x over static batching, because the evaluated workload had uniform output lengths.

Scanned memo: fused attention kernels

IO-aware tiling keeps attention blocks in on-chip SRAM and avoids writing the full attention matrix to HBM. On our A100 test the fused kernel reached a 2.4x speedup over the unfused baseline at sequence length 4096, and attention memory became linear in sequence length.

Recommendation: combine fused attention with 4-bit weights and a paged KV cache for 24 GB GPUs.